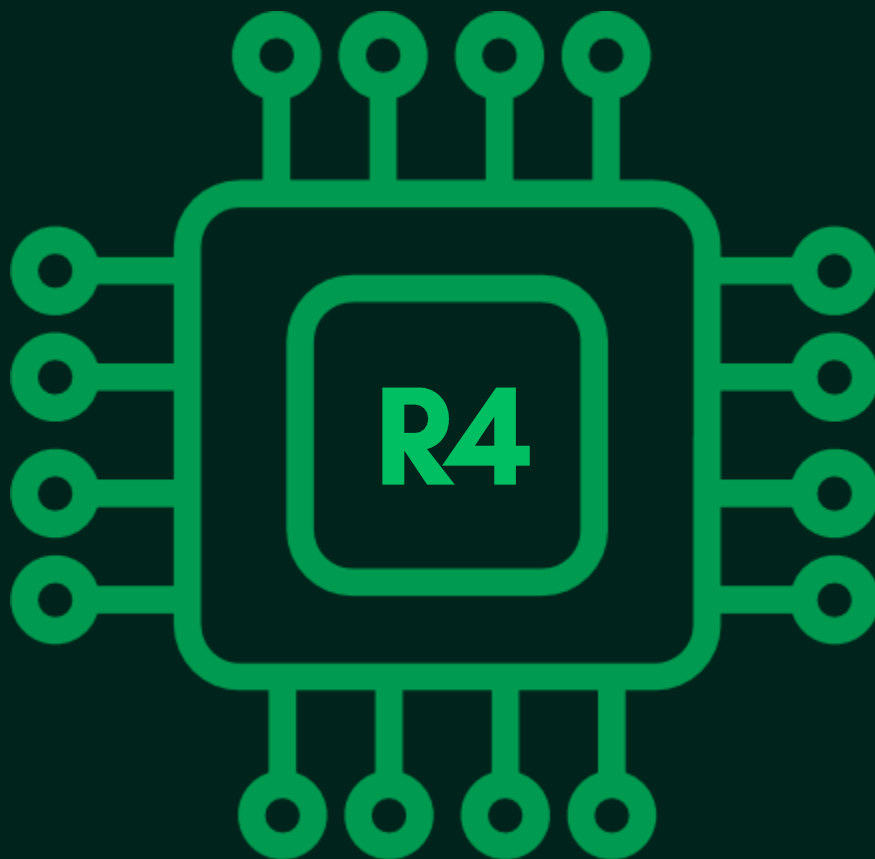


Digital IC Verification Course

BY HASSAN KHALED



Course Description

Dive deep into the essential world of Digital IC Verification with our comprehensive course. This program is meticulously designed to equip you with mastery of SystemVerilog for advanced hardware description and modeling, and UVM (Universal Verification Methodology), the industry-standard framework for building robust and reusable testbenches. You'll gain the practical skills and theoretical understanding necessary to tackle complex verification challenges, significantly boosting your career prospects in the highly in-demand field of ASIC and FPGA design.

What You'll Learn/Key Topics

Course Duration will be 14 sessions divided into 7 main Modules (With Final Projects For each)

- 1] SystemVerilog**
- 2] SystemVerilog Assertions (SVA)**
- 3] Universal Verification Methodology (UVM)**
- 4] Register Abstraction Layer (RAL)**
- 5] Formal Verification**
- 6] Gate Level Simulation (GLS)**
- 7] AI/ML with HW Verification**

SystemVerilog

- Introduction to SystemVerilog
- SV Data Types
- SV Arrays
- Procedural Statements & Flow
- Tasks & Functions
- Object Oriented Programming (OOP)
- Randomization & Constraints (CRT)
- Code Coverage
- Functional Coverage
- Interfaces & clocking blocks
- IPC / ITC

SystemVerilog Assertions

- Introduction to Assertions
- Immediate Assertions
- Concurrent Assertions
- Properties and sequences
- Assertions operators
- Assertions System Functions

- **About UVM**
- **OOP Classes and Extensions**
- **Polymorphism**
- **Fundamentals For Design Patterns**
- **UVM Phases**
- **UVM Reporting**
- **UVM Structure**
- **UVM Object**
- **UVM Component**
- **UVM Test**
- **UVM Environment**
- **UVM Agents**
- **UVM Driver**
- **UVM Monitor**
- **UVM Transactions**
- **UVM Sequence**
- **Configuration DB**
- **UVM TLM**
- **Virtual Sequences**
- **Virtual Sequencers**

UVM RAL

- Introduction to Register Abstraction Layer
- Components of RAL
- UVM RAL Structure
- Register
- Register Fields
- Register Block
- Adapter
- Implicit Predictor
- Explicit Predictor
- Passive Predictor
- Register Access Methods
- API Methods

Formal Verification

- Introduction to Formal
- Property Checking
- SVA for Formal
- Equivalence Checking
- Formal vs Simulation
- Tools & Flows

Gate Level Simulation

- **Introduction to GLS**
- **Netlist Simulation**
- **SDF Back-Annotation**
- **Timing & Functional GLS**
- **X-Propagation**
- **GLS Debug Techniques**
- **GLS vs RTL Delta**

AI/ML with HW Verification

- **Overview on LLMs**
- **LLM-Assisted Debug**
- **Agentic-Based Verification**
- **AI powered Tools**
- **AI/ML Applications with HW**
- **What is Next in HW with AI Era**

Course Schedule

- Course coming round will start July-2026 [Summer Holiday]
- Offline [14 Session/ 2 sessions per week] (Cairo – Nasr City)
- Sessions will be on Weekend [Friday/Saturday]
- Session Period is 3 – 4 hours

How to Enroll

- Firstly, please Apply through my [Website](#)
- Total Fees: **3500 LE** including..
 - Course notes
 - Certificate
- **Payment Method:**
 - Vodafone Cash / InstaPay: **01027471906**
 - U can pay full fees or **1000LE** deposit with the rest of the fees with First Session
 - Once Payment is done please send me on WhatsApp a snapshot with the payment
- **Deadline for Payment:** **26/6/2026**

Notes

- Fees are not refunded after first session.
- The Course Attendance can't be delayed for next rounds.
- Do not forget to join my FB group and WA group for new updates:
 - [FB_Group](#)
 - [WA_Group](#)

Best Of Luck